

Homework for Chap 8

With the effective mass approximation, do the following calculations:

1. Suppose the bandgap of GaAs at $T=300$ K is 1.42 eV and the electron effective mass is $0.062m_0$, calculate the density of states at energy of 1.52 eV and the effective density of states for nondegenerate electron gas in three dimensional (3D) GaAs.
2. For a GaAs quantum well with the well width of 3 nm, suppose the barrier height is infinite, calculate the first quantized energy E_1 for the 2-dimensional (2D) electrons and the corresponding electron density of states at this quantized energy E_1 .
3. Calculate the electron density of states for 3D electron gas at the energy E_1 , obtained in Q2, and compare the value with the density of states for 2D electrons at E_1 .